

- 20 A high electric potential is applied between two electrodes of a hydrogen discharge tube so that the gas is ionized. Electrons then move towards the positive electrode and protons towards the negative electrode. In each second, 5.0×10^{18} electrons and 2.0×10^{18} protons pass a cross-section of the tube. What is the current, in amperes, flowing in the discharge tube?

A 0.16

B 0.48

C 0.80

D 1.1

Ans: D

$$I = I_e + I_p$$

$$= n_e(e) + n_p(p) \quad \text{charge on proton, } p = \text{charge on electron, } e = 1.6 \times 10^{-19} \text{ C}$$

$$= (5.0 \times 10^{18})(1.6 \times 10^{-19}) + (2.0 \times 10^{18})(1.6 \times 10^{-19})$$

$$= 1.1 \text{ A}$$

- 21 A lamp is connected to a power supply of negligible internal resistance. Which circuit could